Present Day, Present Time: AI Exposure & Labor Market Risk

Leyao Wang

University of Virginia
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Setup: Tasks, Workers, and Production

- A continuum of cognitive **tasks** $i \in [0, 1]$, ordered by susceptibility to AI
- **Workers** of type i^* specialize in task i^*; human supply of task i is $f(i)$ — fixed and inelastic
- **CES production** aggregates all tasks into output:

$$Y = \left[\int_0^1 y(i)^{\frac{\sigma-1}{\sigma}} \, di \right]^{\frac{\sigma}{\sigma-1}}, \qquad \sigma > 0$$

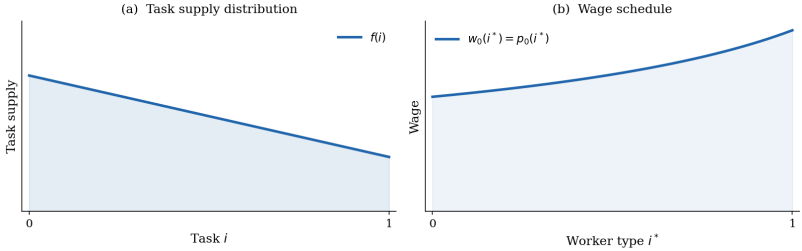
- Output price P_Y fixed (small open economy — think: world price)

Pre-AI Wages

Firms minimize cost → market clearing → each worker earns the competitive price of their task:

$$w_0(i^*) \;=\; P_Y \left(\frac{Y_0}{f(i^*)} \right)^{1/\sigma}$$

Pre-AI Equilibrium



Scarce tasks (high i) pay more; abundant tasks (low i) pay less.

How AI Enters: A Supply Shock

AI adds supply $s(i) \geq 0$ to the tasks it can perform:

$$s'(i) < 0 \text{ for } i < I^*, \qquad s(i) = 0 \text{ for } i \geq I^*$$

- $I^* =$ **AI capability frontier**
- AI is best at routine tasks (low i), contribution shrinks as $i \to I^*$
- Tasks above I^*: beyond AI's reach
- As AI improves, I^* shifts right

Think: offshoring your tasks to Claudeland

Post-AI Wages

Effective supply: $\bar{y}(i) = f(i) + s(i)$. Output expands to $\bar{Y} > Y_0$.

Same CES logic gives the **post-AI wage**:

$$\boxed{w_1(i^*) \;=\; P_Y \left(\frac{\bar{Y}}{f(i^*) + s(i^*)} \right)^{1/\sigma}}$$

Compare to pre-AI: $w_0 = P_Y (Y_0/f)^{1/\sigma}$. Two things moved:

- $\uparrow$ **Numerator** $\bar{Y} > Y_0$: economy grew $\rightarrow$ helps *everyone*
- $\downarrow$ **Denominator** $f+s > f$ for $i < I^*$: more supply $\rightarrow$ hurts exposed workers

Who wins and who loses depends on which effect dominates.

Three Groups of Workers

Above I^*

$s = 0$, denominator unchanged.

$$\frac{w_1}{w_0} = \left(\frac{\bar{Y}}{Y_0}\right)^{1/\sigma} > 1$$

Pure gain from output expansion.

Deep inside $[0, I^*)$

s/f large — denominator grows faster than numerator.

$$\frac{s(i^*)}{f(i^*)} > \frac{\bar{Y}}{Y_0} - 1$$

Wage compression. If below reservation $\rightarrow$ displacement.

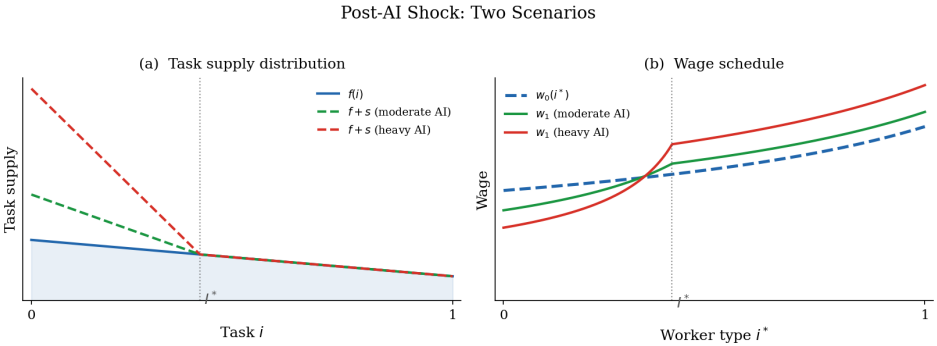
Near I^* from below

$s(i^*) \approx 0$, so denominator barely moved.

$$\frac{w_1}{w_0} \approx \left(\frac{\bar{Y}}{Y_0}\right)^{1/\sigma} > 1$$

Still gain — split is at $\hat{\imath} < I^*$, not at I^*.

Post-AI: Two Scenarios



Left: task supply before and after AI. Right: resulting wages.

Moderate AI $\rightarrow$ mild compression. Heavy AI $\rightarrow$ deep compression below $\hat{\imath}$, larger gains above.

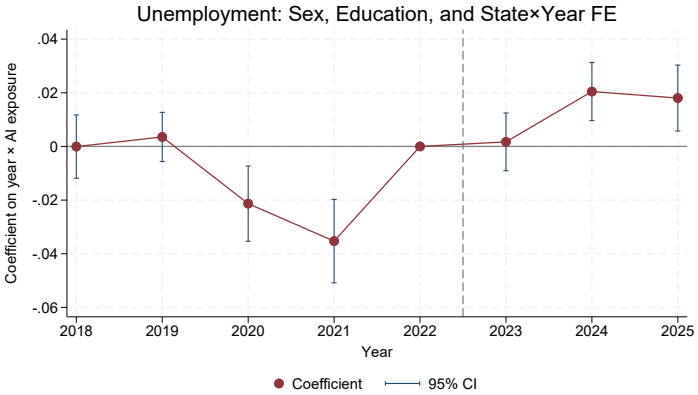
Empirical Strategy

$$Y_{it} = \alpha + \sum_{t \neq 2022} \beta_t \left(\mathbf{1}[t] \times \mathrm{AIExp}_i \right) + X_{it}' \Gamma + \varepsilon_{it}$$

X_{it}: sex, education, state×year FEs
SEs clustered by occupation Baseline: 2022 ($\beta_{2022} \equiv 0$)

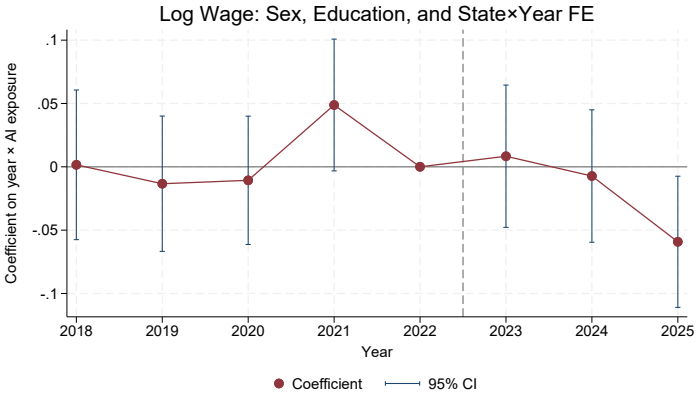
- **Data:** CPS, 500K+ obs, 2018–2025
- **AI exposure:** Eloundou et al. (2024), occ-level LLM score
- **Strategy:** high vs. low exposure, before vs. after ChatGPT
- **Model says:** β_t diverges post-2022

Result 1: Unemployment



- *Interesting* pre-trends — what happened in 2020
- **Post-2022:** rising unemployment gap for high-exposure workers
- Robust across specifications

Result 2: Wages



- Flat pre-trends — clean baseline
- **Post-2022:** wages compress for high-exposure workers
- Non-college workers bear the brunt

Thank you!

Questions welcome.

Bottom line:

AI is like offshoring —
some workers face immediate
wage and employment risk

The data says:

Yes, high-exposure workers
are already seeing it —
unemployment *and* potentially wages

Backup: The Dirty Work

Formal derivations, three-group analysis, and discussion of I^*

Derivation: Pre-AI Equilibrium

Step 1. Cost minimization $\rightarrow$ FOC: $p_0(i) = \mu \cdot (Y/y(i))^{1/\sigma}$, $\quad \mu = P_Y$ in competitive eq.

Step 2. Task demand: $y^d(i) = Y(P_Y/p_0(i))^{\sigma}$

Step 3. Market clearing $y^d(i) = f(i)$:

$$p_0(i) = P_Y \left(\frac{Y}{f(i)} \right)^{1/\sigma}$$

Step 4. Plug $y(i) = f(i)$ into CES $\rightarrow$ output is pinned:

$$\boxed{Y_0 = \left[\int_0^1 f(i)^{(\sigma-1)/\sigma} \, di \right]^{\sigma/(\sigma-1)}}$$

Y_0 depends only on worker distribution f — not on prices.

Derivation: Post-AI Equilibrium

Step 5. AI adds supply $s(i)$. Effective supply: $\bar{y}(i) = f(i) + s(i)$.

Repeat Steps 1–4 with $\bar{y}$ in place of f:

$$\bar{Y} = \left[\int_0^1 (f(i) + s(i))^{(\sigma-1)/\sigma}\, di\right]^{\sigma/(\sigma-1)} > Y_0$$

Post-AI wage:

$$\boxed{w_1(i^*) = P_Y \left(\frac{\bar{Y}}{f(i^*) + s(i^*)}\right)^{1/\sigma}}$$

Wage ratio (Group 2 formula):

$$\frac{w_1(i^*)}{w_0(i^*)} = \underbrace{\left(\frac{\bar{Y}}{Y_0}\right)^{1/\sigma}}_{\text{output expansion}>1} \cdot \underbrace{\left(\frac{f(i^*)}{f(i^*) + s(i^*)}\right)^{1/\sigma}}_{\text{supply effect}\leq 1}$$

Three-Group Analysis (Formal)

Group 1 ($i^* \geq I^*$, $s = 0$): $w_1/w_0 = (\bar{Y}/Y_0)^{1/\sigma} > 1$ — uniform gain, increasing in aggregate AI supply.

Group 2 ($i^* < I^*$, s/f large): Compression whenever $s(i^*)/f(i^*) > \bar{Y}/Y_0 - 1$.
Because $\bar{Y}/Y_0$ is a CES average over *all* tasks (most with $s = 0$), while s/f is concentrated on AI-supplied tasks, this condition holds for heavily exposed workers.

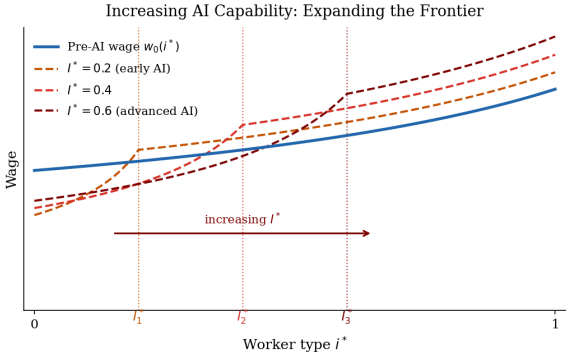
Group 3 ($i^* < I^*$, s/f small): Near I^*, $s \to 0$ so supply effect ≈ 1 and output expansion dominates $\to$ these workers gain despite $i^* < I^*$.

Threshold $\hat{\imath} < I^*$**:** $s(\hat{\imath})/f(\hat{\imath}) = \bar{Y}/Y_0 - 1$.
Winners: $i^* > \hat{\imath}$. Losers: $i^* < \hat{\imath}$. The boundary lies *strictly below* I^*.

Further Discussion of I^*

As AI improves, I^* shifts right

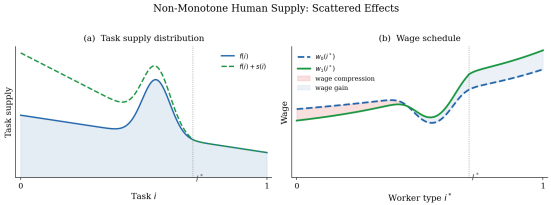
More workers exposed, deeper compression, $\hat{\imath}$ shifts rightward.



Each curve: w_1 for a different I^*. Higher I^* $\longrightarrow$ wider compression.

Non-monotone $f(i)$: scattered effects

If many workers cluster in a mid-complexity task, the local s/f ratio drops, creating pockets where workers gain even inside $[0, I^*)$.



Bump in f near $i \approx 0.55$: local dip in w_0, gain pocket inside the frontier.